

ML Mini Project · PES University

COVID-19 Risk Estimation

Predicting ICU admission, intubation and mortality from baseline patient data

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Why early triage matters

During surges, hospitals run short of ICU beds and ventilators. A risk score at admission helps decide who needs them first.

ICU admission

Who needs intensive care

Intubation

Who needs a ventilator

Mortality

Who is at risk of death

Dataset

23,465

confirmed COVID-19 patients

11

input features

80/20

train / test split

Inputs known at admission: age, gender, pneumonia and eight comorbidities including diabetes, COPD, hypertension and obesity.

Source: Mexican Ministry of Health COVID-19 dataset (Kaggle)

Approach

4.2%

need ICU or intubation (9.2% die)

Predicting "no" for everyone scores 95.8% accuracy and misses every critical patient.

So we use balanced class weights and prioritise recall.

**Logistic Regression
(deployed)**

SVM

Decision Tree

Random Forest

Logistic regression results

Outcome	Recall	Precision	ROC-AUC
ICU admission	0.84	0.12	0.829
Intubation	0.94	0.14	0.877
Mortality	0.80	0.25	0.848

Held-out test set, n = 4,693. Precision is low by design: we accept false alarms to catch critical patients.

SVM results

Outcome	Recall	Precision	ROC-AUC
ICU admission	0.82	0.12	0.802
Intubation	0.92	0.14	0.846
Mortality	0.83	0.23	0.811

Held-out test set, n = 4,693. Balanced class weights, same features and split.

Decision tree results

Outcome	Recall	Precision	ROC-AUC
ICU admission	0.83	0.12	0.830
Intubation	0.90	0.13	0.852
Mortality	0.81	0.23	0.839

Held-out test set, n = 4,693. Balanced class weights, same features and split.

Random forest results

Outcome	Recall	Precision	ROC-AUC
ICU admission	0.83	0.12	0.830
Intubation	0.90	0.13	0.852
Mortality	0.81	0.23	0.839

Held-out test set, n = 4,693. Balanced class weights, same features and split.

Why logistic regression

ROC-AUC	ICU	Intubation	Mortality
Logistic Regression	0.829	0.877	0.848
SVM	0.802	0.846	0.811
Decision Tree	0.830	0.852	0.839
Random Forest	0.830	0.852	0.839

The results are clustered, and logistic regression gives the best results while being simpler than the others. So we chose it.

Strongest risk factor

43.8x

higher odds of intubation with pneumonia

- Pneumonia dominates all three outcomes
- Age is the next driver of mortality
- Obesity, diabetes and immunosuppression add risk

Live demo

demo.py

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streamlit run demo.py
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- Enter age, gender and conditions
- Click Predict
- Read ICU, intubation and death risk with flags

Precision vs recall in medicine

LOW RECALL

A critical patient is missed

Care is delayed and the patient may die. This cost is severe and cannot be undone.

LOW PRECISION

A stable patient is flagged

The patient gets extra monitoring. This cost is small and easy to recover from.

Only 4.2% of patients need ICU or intubation, so 12% precision is about 3× better than chance, while catching 84–94% of them.